

Can One Volatility-Manage Fixed-Income Portfolios?

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Abstract

We present an extension to the US fixed-income domain of the volatility-managed portfolio construction introduced by Moreira and Muir (2017) in the equity context. We show that with US Treasuries the most effective indicators of when money should be taken off the table are changes in the MOVE index (ie, in implied volatilities). We recover the positive α found in equities, and we show that, for every maturity, the volatility-managed portfolios have Sharpe ratios significantly higher than the corresponding buy-and-hold portfolios. When the changes in the MOVE index are transformed to Gaussian equivalents, Sharpe ratios increase, α become more positive and significant, and drawdowns and leverage decrease. The strategy still works if a zero floor is imposed on the notional, to reflect the constraints of long-only managers.

Key Words: Volatility-managed portfolios, US Treasuries, MOVE index.

1 Introduction

In a widely quoted paper, Moreira and Muir (2017) found that ‘[m]anaged portfolios that take less risk when volatility is high, produce large alphas, increase Sharpe ratios, and produce large utility gains for mean-variance investors’. As the authors point out, their proposed strategy sits awkwardly with asset pricing theory because it takes relatively less risk in recessions. As they say, ‘[t]his rules out typical risk-based explanations and is a challenge to structural models of time-varying expected returns’. Their study, and the work that followed in its wake, focussed on equity returns. In this paper we turn our attention to fixed-income markets, and we ask whether volatility management can work also for US Treasuries. We show that taking money off the table when volatility increases can be highly effective also in fixed-income instruments (US Treasuries), but, if anything, our finding deepens the puzzle: until the recent inflationary bout, markets have relied on the deployment of the ‘Greenspan put’ – on the expectation, that is, that central banks would cut rates in periods of distress. Reducing one’s exposure to fixed instruments just when the going gets tough would therefore sound like a recipe to sit out the periods when Treasuries are expected to post the highest gains. Yet, we find that this is indeed the case. As we discuss in Section 5, this raises important and unanswered questions about the effectiveness of the reliance of market participants on the Greenspan put during the 2011-2022 period – a period which is fully covered in our analysis.

Both Moreira and Muir (2017)’s and our strategy are special cases of timing strategies. Timing strategies can be seen as prescriptions of how to modulate the constant notional of a buy-and-hold on the basis of some signal. What characterizes timing strategies is therefore the state dependent notional, $N(\omega)$, that dictates in which states of the world, ω , the exposure to the underlying should be increased, decreased, or perhaps even reversed – the only constraint on the ‘rule’ $N(\omega)$ is that it should not suffer from peek-ahead bias. (A bit more technically, the process $N(\omega)$ should be non-anticipating, ie, adapted to the filtration generated by the observable market variables, which in our case are option and bond prices.)

We show in Section 3 that *any* such non-anticipating strategy has an associated α , which is given the covariance between the expected return of the buy-and-hold portfolio and the time-varying notional. (Note that to calculate this α we do not need the difficult-to-estimate first moment of the expected return, but only its variance and correlation with the proposed notional.)

When we look at the problem from this perspective, the search for timing α therefore boils down to the search for what in fixed-income parlance are referred to as ‘return-predicting factors’ – quantities, that is, that covary positively with the expected returns. In this paper we identify one such quantity: we document in Section 2 that there is a statistically very significant inverse relationship between the returns of bonds of any maturity and changes in the MOVE index. Since the latter is a combination of the implied volatilities of short-dated Treasuries options, this means that the notional of the dynamic strategy should be reduced when the implied volatility increases. So, indeed, as pointed out above, our timing strategy suggests to reduce exposure just when volatility increases – hence the paradox that Moreira and Muir (2017) identified in the equity arena, and that we find in the fixed-income world. Since in both asset classes it is volatility (albeit in our case implied rather than realized) that determines the notional, we still call our strategy ‘volatility managed’.

As for the choice of *changes* in the implied volatilities of a basket of Treasury options as our scaling indicator, we point to the work by Wang and Yan (2020), who ‘show that strategies scaled by downside volatility exhibit significantly better performance than strategies scaled by total volatility’. If high volatility is associated with a higher probability of downward jumps (as documented in Patton and Shepperd (2015)), then by using changes in (implied) volatility to manage the portfolio notional we recover the intuition of the findings by Wang and Yan (2020).

We show in Section 4 that the α of our strategy is positive for all maturities, and that its Sharpe ratio is significantly higher than the Sharpe ratio of the buy-and-hold strategy. The leverage that it requires is similar to what found by Moreira and Muir (2017). Reducing the leverage by capping the absolute value of the notional reduces the α and the difference in the Sharpe Ratios, but they remain statistically significant. We also note that we obtain our results for a timing strategy that allows for long and short positions in the underlying bonds (the Moreira and Muir (2017) prescription generates strictly positive notionals). For asset managers with a long-only mandate, our strategy remains profitable when the notional is floored at zero, but, again, the α and the difference in the Sharpe Ratios are reduced.

When we examine the distribution of implied volatility changes, we find that, for all maturities, it is far more leptokurtic than the distribution of Treasury returns. In the construction of our trading signal we therefore also use transformed variables that can be thought of as the percentile-by-percentile Gaussian equivalents of the implied volatility changes. When we do so, we find that the Sharpe ratios improve significantly, and that the leverage decreases strongly, making our proposed approach even more attractive.

Finally, we note that, as in the case of the strategy by Moreira and Muir (2017), there is an undetermined scaling constant in our strategy. For Moreira and Muir (2017) it has exactly no effect on the Sharpe Ratio.¹ In our case, the undetermined constant does have an effect on the the Sharpe Ratio, but it is very muted. As Moreira and Muir (2017), we recommend to choose the scaling constant in such a way as to have the same portfolio volatility as the buy-and-hold portfolio. We acknowledge, however, the criticisms that have been levelled at the Moreira and Muir (2017) recommendation in this respect (the constant is only exactly known *ex post*), and that this issue has not been resolved in a completely satisfactory manner.

¹This is exactly true for absolute (not percentage) returns. See the discussion in Guo and Liu (2019) in this respect.

2 Data Description and Analysis

For our analysis we use Treasury bonds and the Merrill Option Volatility Estimate (MOVE) index (described below). The continuously compounded zero-coupon bond yields of maturities from 1 to 30 years were sourced from the Board of Governors of the Federal Reserve System database (Guarkaynak, Sachs, and Wright (2007)). These yields were obtained from actual Treasury prices using (modifications of) the Nelson and Siegel (1987) model.

The MOVE index tracks the implied volatility of a basket of 1-month options on 2-year, 5-year, 10-year and 30-year Treasuries. At time t , for each option the implied normal yield volatility, $\sigma^{impl}(t, T)$, for maturity T and premium $C(t)$ is given by

$$\sigma^{impl}(t, T) = \sqrt{\frac{2\pi}{T}} C(t) \quad (1)$$

These implied volatilities are then aggregated as follows:

$$MOVE(t) = \sum_{i=1}^4 w_i \cdot \sigma^{impl}(t, T_i) \cdot 10,000 \quad (2)$$

with $w_1 = 0.2$, $w_2 = 0.2$, $w_3 = 0.4$ and $w_4 = 0.2$. The multiplication factor 10,000 converts to basis points. Roughly speaking, the MOVE index reflect the annualized average option-implied expected volatility for a basket of options on Treasuries, expressed in basis points.

Because of data availability, for our analysis we used daily realizations of the yield and MOVE data from 12-Nov-2002 to 22-May-2026. This period covers a number of very important events, including the 2008 financial crisis, the Great Recession (2009-2001), the COVID crisis (2020), the ensuing inflation increase (2022), and the beginning of the Iraq War (2026). The time series of the MOVE index and of the 30 yields used in the analysis are shown in Fig 1.

From the yields, y_t^T , we build bond prices, P_t^T , as

$$P_t^T = \exp(-y_t^T \cdot (T - t)) \quad (3)$$

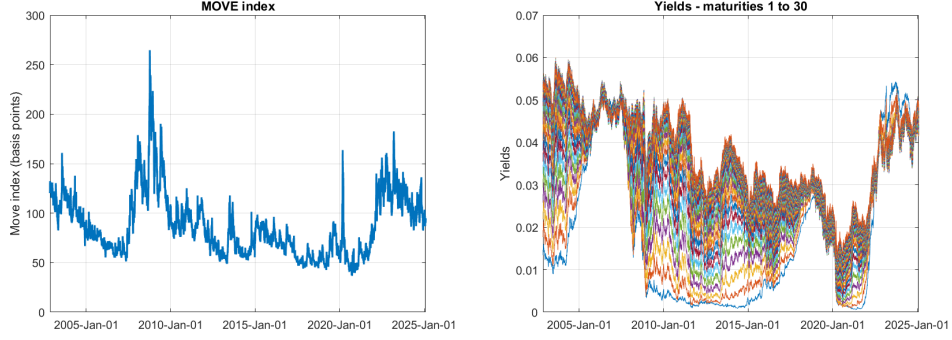


Figure 1: The time series of the MOVE index (left panel) and of the 30 yields (right panels) used in the analysis.

From these we obtain (absolute or percentage) returns, $R^T(t, t+1)$, as

$$\begin{aligned} R_{abs}^T(t, t+1) &= P_{t+dt}^{T-dt} - P_t^T \\ R_{perc}^T(t, t+1) &= \frac{P_{t+dt}^{T-dt} - P_t^T}{P_t^T} \end{aligned} \quad (4)$$

with dt equal to one business day.² From our database the yields of maturity $T - dt$ are not known. We therefore linearly interpolated the yields between T and $T - 1$. The effect is minimal because $dt \cong \frac{1}{252}$, and almost identical results are obtained if one sets $y_{t+dt}^{T-dt} = y_{t+dt}^T$.

Given these returns (absolute or percentage), we ran the following regressions:

$$R^T(t, t+1) = \alpha^T + \beta^T \cdot \Delta MOVE(t-1, t) + \epsilon_t^T \quad (5)$$

where $\Delta MOVE(t-1, t)$ is the change in the MOVE index from time $t-1$ to time t (and therefore known at time t), and $R^T(t, t+1)$ is the return of the zero-discount bond of maturity T from time t to time $t+1$. Note that the choice of time indexing ensures no peek-ahead bias in the estimation of the regression coefficients: at time t the change in the MOVE index from time $t-1$ to time t is known, and we make use of this information to

²Guo and Liu (2019) discuss the difference between using simple or percentage returns in the context of Moreira and Muir (2017)'s work, who use monthly holding periods. Since we are using daily returns, the difference is in our case minimal.

determine the notional that applies from time t to time $t + 1$.

The average absolute change in the MOVE index is 2.73, with the 5th, 10th, 90th and 95th percentiles of the change distribution equal to -6.10 , -4.04 , $+4.34$ and $+6.73$, respectively. We fitted the changes in the MOVE index to a Student- t distribution, and we obtained by MLE $\mu = -0.196505[-0.270621, -0.122389]$, $\sigma = 2.22159[2.13578, 2.31085]$ and $nu = 2.26828[2.09524, 2.45562]$ (the numbers in square brackets display the 95 percent confidence band for each coefficient). By visual inspection (see the left panel of Fig 2) the fit seems to be good. However the Q-Q plot in the right panel highlights the extremely leptokurtotic nature of the empirical distribution. Since the bond returns do not display such pronounced fat tails, this suggests a challenge for the regression in Eq 5. We have therefore also calculated transformed MOVE changes, by building the empirical cumulative distribution of the MOVE changes, and then applying the inverse standard cumulative distribution. Symbolically, we obtain transformed variables, $T\Delta MOVE$, given by

$$T\Delta MOVE = N^{-1}(EC(\Delta MOVE)) \quad (6)$$

where N^{-1} denotes the inverse cumulative of the Gaussian distribution, and $EC(\Delta MOVE)$ the empirical cumulative distribution of $\Delta MOVE_t$. This transformation improves all the performance metrics that we present in Section 4. For conservatism, we focus the presentation on the non-transformed-data analysis, and present the comparison in Section 4.1.

Despite this problem, Fig 3 shows that, even with the non-transformed data, all the slope coefficients are statistically significant at the 90% confidence level, and the first 21 maturities (out of 30) are significant at the 95% confidence level. For all maturities, the β^T coefficients are negative, and they imply a decrease in (annualized) returns – a decrease that, for the 5-year maturity, can be as large as 1.25% for a unit change in the MOVE coefficient. In the light of the discussion in the introductory section, the fact that expected returns are inversely related to changes in the MOVE index inspires the volatility-management strategy that we discuss in Section 3.

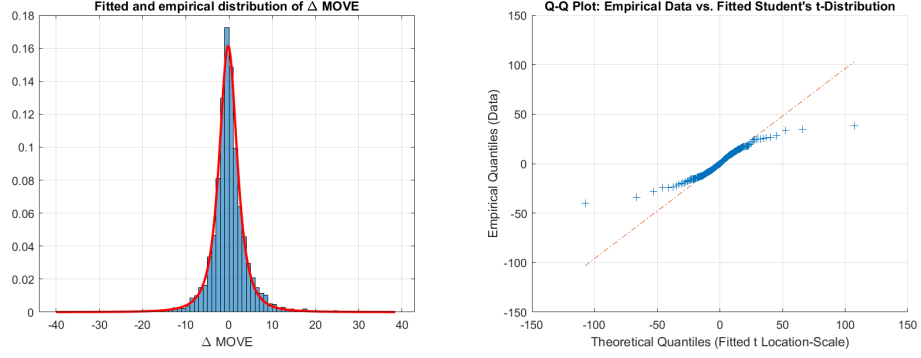


Figure 2: Left panel: The normalized histogram of realized $\Delta MOVE$ (blue bars) and the fitted Student- t distribution (red curve). Right panel: Q-Q plot for the theoretical distribution (x axis) and the empirical data (y axis).

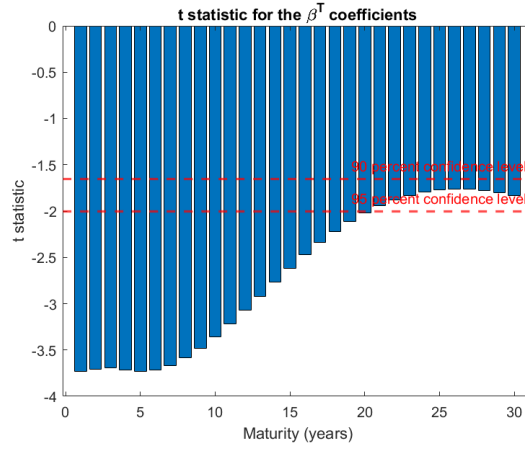


Figure 3: The t statistic for the slope coefficients, β^T , as a function of bond maturity (on the x axis). The horizontal red lines show the 95% and 90% confidence levels.

3 The Methodology

Consider a reference portfolio, Π_t^T , that holds at time t one unit of the T maturity bond. Consider also a variable-notional portfolio, $\hat{\Pi}_t^T$, that holds at time t N_t units of the same T maturity bond. For the moment we leave the notional N_t unspecified, and we then recover

both the Moreira and Muir (2017) approach and more general specifications.

Following closely Moreira and Muir (2017), we assume that the return from the reference portfolio is given by

$$\frac{d\Pi_t^T}{\Pi_t^T} = (\mu_t^T - r_t) dt + \sigma_t^T \cdot dz_t^T \quad (7)$$

The return from the notional-adjusted portfolio can then be written as

$$\frac{d\hat{\Pi}_t^T}{\hat{\Pi}_t^T} = r_t \cdot dt + N_t \cdot \left(\frac{d\Pi_t^T}{\Pi_t^T} - r_t \cdot dt \right) \quad (8)$$

Therefore, taking expectations, we have

$$\mathbb{E} \left[\frac{d\hat{\Pi}_t^T}{\hat{\Pi}_t^T} - r_t \cdot dt \right] = \mathbb{E} \left[N_t \cdot \left(\frac{d\Pi_t^T}{\Pi_t^T} - r_t \cdot dt \right) \right] = \mathbb{E} [N_t \cdot \mu_t] \cdot dt \quad (9)$$

Let's now regress the returns from the notional-adjusted portfolio against the returns of the reference portfolio:

$$\frac{d\hat{\Pi}_t^T}{\hat{\Pi}_t^T} = \alpha^T + \beta^T \cdot \frac{d\Pi_t^T}{\Pi_t^T} + \epsilon_t^T \quad (10)$$

from which we obtain

$$\alpha^T = \mathbb{E} \left[\frac{d\hat{\Pi}_t^T}{\hat{\Pi}_t^T} \right] - \beta^T \cdot \mathbb{E} \left[\frac{d\Pi_t^T}{\Pi_t^T} \right] \quad (11)$$

Since

$$\mathbb{E} [N_t \cdot \mu_t] = \mathbb{E} [N_t] \mathbb{E} [\mu_t^T] + \mathbb{COV} [N_t \cdot \mu_t] \quad (12)$$

and

$$\beta = \mathbb{E} [N_t] , \quad (13)$$

by substitution one obtains for the (annualized) α of the notional-adjusted strategy

$$\alpha^T = \mathbb{COV} [\mu_t^T, N_t] \quad (14)$$

We note that, in order to estimate the expected profitability of the strategy we do not need to know the difficult-to-estimate expected return of the reference portfolio, but only how it covaries with the notional.

The first (trivial) observation is that, if there is no correlation between the expected return of the reference portfolio and the time-varying notional, then the alpha will be zero. Next, we note that the Moreira and Muir (2017) specification for the notional is

$$N_T^{MM} = \frac{c}{\sigma_t^2} \quad (15)$$

where c is a normalization constant that does not affect the Sharpe Ratio, and σ_t^2 is the conditional variance at time t . If returns are high when the conditional variance is low, then the alpha will be positive. Our specification for the notional is

$$N_t^{MOVE} = -c \cdot \Delta MOVE_t \quad (16)$$

where c is different scaling constant, and $\Delta MOVE_t$ is the time- t change in the MOVE index. We have shown in Section 2 that the correlation between returns and changes in the MOVE index is negative, and therefore we can also expect a positive alpha. To see how large, we move to the empirical analysis in Section 4.

4 Results

In this section we compare the performance of a reference (buy-and-hold) strategy, and of a volatility-managed strategy in which the notional is given by Eq. 16:

$$N_t^{MOVE} = -c \cdot \Delta MOVE_t \quad (17)$$

We assess the relative performance of the two strategies by calculating their Sharpe Ratios, and, following Moreira and Muir (2017), by the intercept α in the regression in Eq 10.

To begin with, Fig 4 shows the cumulative profits for the buy-and-hold strategy (left panel) and for the volatility-managed strategy (left panel) for the 30 maturities. In particular, we note how from 2010 to the onset of the COVID crisis there is a steady accumulation of profits for the volatility-managed strategy, while the buy-and-hold strategy displays a very erratic behaviour.

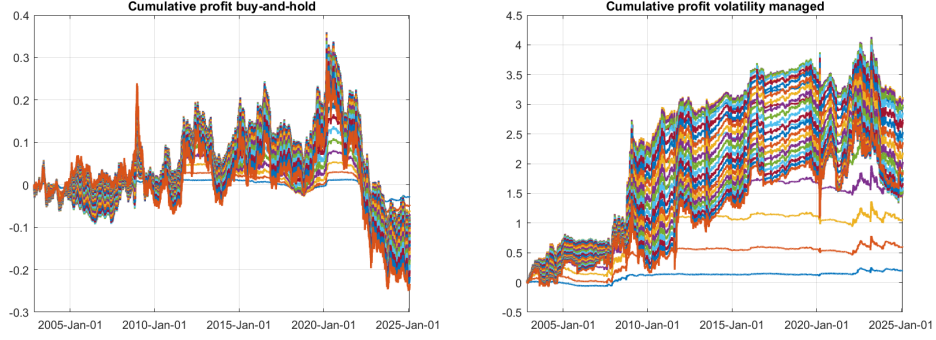


Figure 4: Left panel: The cumulative profit for the buy-and-hold strategy for the 30 maturities . Right panel: The cumulative profit for the volatility-managed strategy for the 30 maturities ($c = 1$).

The differences in Sharpe Ratios between the volatility-managed and the buy-and-hold strategies are shown in Fig 5. Lo's Asymptotic Test (Lo (2019)) for the difference between two Sharpe Ratios rejected for all maturities the hypothesis that the two Sharpe Ratios were not different at the 99.9 confidence level or higher. Since Lo's test assumes that returns are IID and normally distributed, we also implemented the non-parametric Ledoit-Wolf Robust Bootstrap Test (Ledoit and Wolf (2008)) with 1,000 bootstrapping replications and block size equal to 200. The equal-Sharpe Ratio hypothesis with this more robust test was rejected for all maturities at the 95 percent confidence level, and for 26 out of 30 maturities at the 99 percent level (the test marginally failed for maturities 18, 20, 21 and 22).

We next move to the analysis of the maturity-dependent intercepts, α^T , in Eq 10, which we reproduce below for ease of comparison

$$\frac{d\hat{\Pi}_t^T}{\hat{\Pi}_t^T} = \alpha^T + \beta^T \cdot \frac{d\Pi_t^T}{\Pi_t^T} + \epsilon_t^T \quad (18)$$

(This, by the way, is the main metric used by Moreira and Muir (2017) in their analysis.) As Fig 6 shows, the intercepts are positive for all maturities, and are statistically significant at the 90 percent confidence level for 12 out of 30 maturities, and at the 95 percent confidence level for 7 out of 30 maturities (the most significant α s are for maturities from 4 to 11 years). In sum: by varying the notional according to Eq 16, we have created volatility-managed

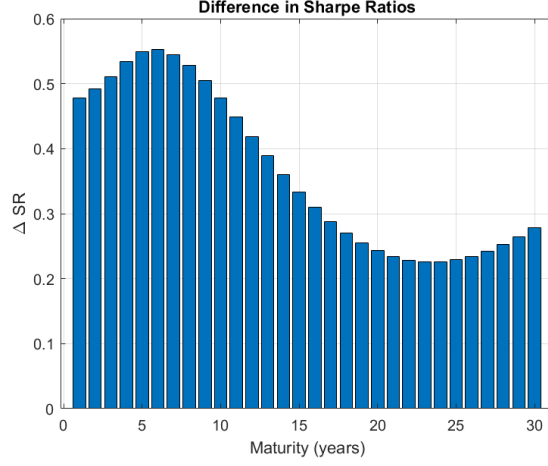


Figure 5: The difference in Sharpe ratios of the volatility-managed strategy against the buy-and-hold strategy.

portfolio that, for each maturity, outperform buy-and-hold portfolios in expected return terms, and offer more attractive Sharpe Ratios.

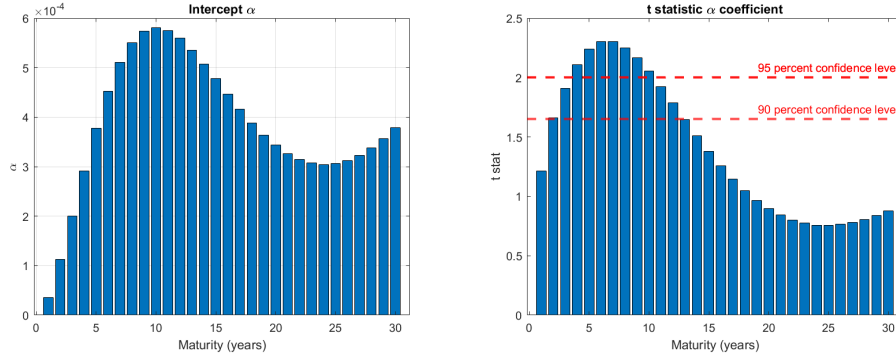


Figure 6: Left panel: The intercept α in the regression $\frac{d\hat{\Pi}_t^T}{\hat{\Pi}_t^T} = \alpha^T + \beta^T \cdot \frac{d\Pi_t^T}{\Pi_t^T} + \epsilon_t^T$. Right panel: the t statistic for the intercept α^T .

We have run several robustness tests. In particular,

1. we have varied the constant c between 0.5 and 3 – we found that the differences in Sharpe Ratios were not statistically significantly different from zero. Clearly, the

leverage increased with c .

2. Since the work by Moreira and Muir (2017) used *variance* rather than volatility, we allowed the state-dependent notional to be given by

$$N_t^{TW} = -c \cdot \Delta(MOVE_t)^\gamma \quad (19)$$

with $\gamma = 0.5, 1, 2$. Again, statistically insignificant or marginally significant variations in the performance metrics were observed. (The best results were obtained with $\gamma = 1$.)

3. We regressed returns not only on changes in the MOVE index, but also on the level of the MOVE index and on an interaction term given by $MOVE_T \cdot \Delta MOVE_t$ to capture the possibility that changes in the index may be more significant for return prediction when the index itself is high or low:

$$\begin{aligned} R^T(t, t+1) = & \alpha^T + \\ & \beta_1^T \cdot \Delta MOVE(t-1, t) + \\ & \beta_2^T \cdot MOVE(t-1, t) + \\ & \beta_3^T \cdot (\Delta MOVE(t-1, t) \cdot MOVE(t-1, t)) + \\ & \epsilon_t^T \end{aligned} \quad (20)$$

None of the coefficients β_2^T and β_3^T were found to be statistically significant.

4. We ran the analyses above using the VIX rather than the MOVE index. Neither levels, changes or interaction terms between levels and changes proved statistically significant.

4.1 Using Transformed Data

As anticipated in Section 2, all the performance metrics improve, sometimes substantially, when we use Gaussian-transformed changes in the MOVE index. Intuitively, the Gaussian transformation trims the extremes of the tails of the MOVE changes distribution, and,

by doing so, produces variables that can better explain in a linear regression the non-particularly leptokurtic returns. In particular, we found the following.

1. The leverage implied by the notional obtained as $N_t^{TW} = -c \cdot T\Delta MOVE_t$ is significantly reduced: for $c = 1$, with non-transformed variables the 5th, 10th, 90th and 95th percentiles of the notional were -6.10 , -4.04 , $+4.3$, and $+6.73$; with transformed variables the same percentiles were -0.64 , -0.28 , $+2.28$ and $+2.65$. The ‘unreasonable leverage’ that Cederburg, O’Doherty, Wang, and X. S (2020) claim plagues the Moreira and Muir (2017) strategy is no longer encountered.
2. The intercepts, α , are still all positive, and are now significant at the 90 percent confidence level for 14 (*cf* 12) maturities and for 10 (*cf* 7) maturities at the 90 percent confidence level.
3. The differences in Sharpe Ratios between the buy-and-hold and the volatility-adjusted strategies are statistically significantly different from zero (Ledoit-Wolf test) for all maturities at the 90 percent confidence level and for 28 out of 30 maturities at the 95 percent confidence level.

Operationally, under the reasonable assumption that the unconditional distribution of MOVE changes in time invariant, one can build an empirical distribution of raw changes in the MOVE index, and then create an equivalent Gaussian realization by using Eq 6.

4.2 The Long-Only Case

Some portfolio managers face long-only constraints. We have therefore run the same tests presented above, but flooring the notional at 0. When we use the Gaussian-transformed changes in MOVE, we find that the differences in Sharpe ratio between the buy-and-hold and the volatility-adjusted strategy declines marginally, but remains statistically significantly different from zero, and that the statistical significance of the α s is, if anything, increased. The good performance of the zero-floored volatility-managed portfolio with the Gaussian-transformed changes in MOVE comes to little surprise when we recall the low level of

leverage of the 5th and 10th percentiles of the distribution of notionals (-0.65 and -0.28 , respectively).

5 Discussion and Conclusions

The approach we have presented in this work can be seen as an adaptation to the fixed-income world of the volatility-managed portfolio construction suggested by Moreira and Muir (2017) in the equity context. We have shown that in both cases the positive α generated by the volatility-managed strategies stems from the covariance of the state-dependent notional with the return process of the buy-and-hold portfolio. In both cases money is ‘taken off the table’ in periods of expected turbulence. The key difference is that in Moreira and Muir (2017)’s case the expected next-period variance (proxied by the latest realized variance) is the stand-in for market turmoil. In our case this role is played by *changes* in the MOVE index. As Cederburg, O’Doherty, Wang, and X. S (2020) point out, ‘the effectiveness of a volatility-managed portfolio depends upon the relation between lagged volatility and future expected returns, and this risk-return trade-off could be unstable over time’. Since the MOVE index, by virtue of being a weighted combination of short-term implied volatilities, is intrinsically forward-looking, our metric is an attractive candidate because it exactly embeds changes in expected future volatility. Indeed, the work by Christensen and Prabhala (2020) indicate that, at least in the case of the S&P500 index, ‘implied volatility outperforms past volatility in forecasting future volatility and even subsumes the information content of past volatility in some of [their] specifications’. In their study they find that the implied-volatility predictions are biased, but we note that, since we use *changes* in implied volatility, the bias detected by Christensen and Prabhala (2020) to first order cancels out.

In the light of these observations, we have tested against buy-and-hold portfolios the performance of volatility-managed portfolios in which the notional is an inverse function of changes in the implied-volatility MOVE index. We have shown that, by so doing, for all ma-

turities one can indeed build portfolios with positive α s, and more attractive Sharpe Ratios than with a same-maturity buy-and-hold portfolio – especially so, if the the Gaussian-transformed $T\Delta MOVE_t$ changes are used in the portfolio construction.

The portfolio construction by Moreira and Muir (2017) has been criticized because i) of the high leverage required, ii) of the associated large drawdowns (see, eg, Guo and Liu (2019)) associated with the strategy, iii) of the transaction costs, and iv) of the fact that the ‘correct’ value for the constant c can only be known *ex post*. How does our strategy compare in these regards? As for the last objection, one can point out that, while an exact match of the buy-and-hold and volatility-managed portfolio would indeed require knowledge that is only available *ex post*, a reasonable approximation to the volatility of the volatility-managed portfolio can be obtained via back-testing, bootstrapping or, as indeed Guo and Liu (2019) suggest, simulations. Regarding the leverage, while with raw MOVE changes our strategy it can indeed be as high as in Moreira and Muir (2017), when the notional is determined using the Gaussian-transformed quantities leverage (and especially negative leverage) becomes much smaller. And as for the drawdowns, when the notional is made to depend on the Gaussian-transformed $T\Delta MOVE_t$ quantities, we find that, maturity-by-maturity, they are smaller than for the corresponding buy-and-hold drawdowns.

These positive results leave at least one big unanswered question. As we reported, when we tried to use the VIX index and/or its changes to modulate the notional of the volatility-managed portfolios, we found no improvement, and sometimes a deterioration, in the performance over the buy-and-hold portfolio. In a way, this is not surprising, because the MOVE index is exactly tailored to the fixed-income market. However, this finding has broader implications: during the period of the ‘Greenspan put’ (which became the Bernanke put, the Yellen put and, briefly, the Powell put, and which only ended with the inflationary bout of 2022), it is widely held that portfolio managers tended to hedge the downside risk of their equity portfolios with large positions in Treasuries. From this perspective,

the extremely low US and EUR yields in the 2011-2013 period were interpreted not as expectation of the future path of the short rate, but as the insurance premium investors were prepared to pay to hedge their downward equity risk. However, for this strategy to be effective, Treasuries would need to be responsive to ‘equity volatility’ (which is captured by the VIX, not by the MOVE). As mentioned above, we found during this critical period no dependence of Treasury returns on equity risk as proxied by the VIX index. What is more troubling is that, if one wants to interpret the MOVE index as an indicator of broad, and not just fixed-income, turmoil, we found the dependence of the fixed-income returns on the MOVE turmoil index to have the wrong sign for Treasuries to act as effective equity hedges. We are not aware of systematic studies of the actual hedging effectiveness of the ‘Greenspan put’, but, in the light of our findings, this is a topic that would deserve further investigation. In the meantime, the paradox with which Moreira and Muir (2017) open their paper reappears with even greater cogency with the fixed-income instruments that we have analyzed in this study.

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